# Wildfire Detection from Sentinel-2 Sequences Around FIRMS Hotspots

A **split-safe** wildfire screening pipeline · temporal modeling · uncertainty estimation · reproducible experiment tracking

NASA FIRMS · SENTINEL-2 · RESNET-18 · MONTE CARLO DROPOUT · ECE CALIBRATION

| 206 | 3 | 0.9854 | 0.0385 | 2 |
|---|---|---|---|---|
| USABLE SEQUENCES | FRAMES PER SEQUENCE | BEST PR-AUC (TEMPORAL) | BEST ECE (TEMPORAL) | MODELS COMPARED |

## WHY THIS MATTERS

Wildfires spread fast, and early detection matters. Satellite imagery offers broad coverage, but real alerts are **noisy, sparse, and hard to trust**. We want a system that screens candidate hotspots automatically and gives useful confidence estimates — not just a binary answer.

## WHY NOT EXISTING METHODS?

Rule-based alerts are brittle and don't generalize. Single-image classifiers **ignore how fire evolves over time**. Many systems report raw scores without calibration, making them hard to use in practice. Our goal: a pipeline that is both accurate and trustworthy.

> Event-level grouping and split-safe metadata turn noisy hotspot detections into a clean wildfire learning problem.

## DATA

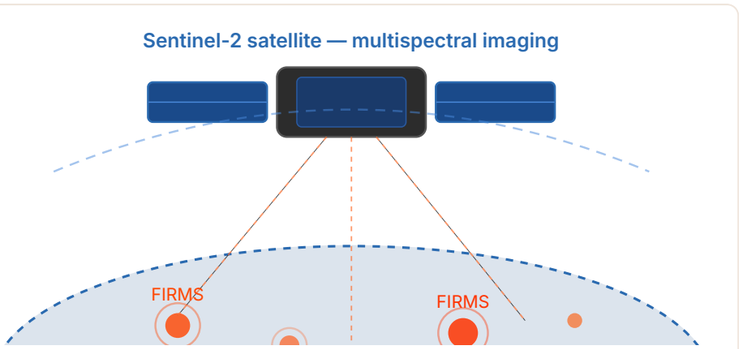
We start from **NASA FIRMS VIIRS** active-fire detections and group nearby detections into wildfire events. For each event we extract short Sentinel-2 sequences centered on the hotspot at three time offsets. We also generate **hard negatives** from nearby confusing locations so the task is not artificially easy. Splits are assigned at the event level to prevent leakage.

PIPELINE INPUT
FIRMS VIIRS detections

SEQUENCE LENGTH
3 frames

FRAME OFFSETS
$t-10 \cdot t-5 \cdot t_0$

USABLE COVERAGE
206 sequences




*Sentinel-2 multispectral satellite captures 3-frame sequences over NASA FIRMS VIIRS active-fire hotspot detections for supervised wildfire classification.*

**206**

sequences — default usable coverage after aggressive extraction

## OUR APPROACH

We compare **two models**. The static baseline uses only the final frame with a ResNet-18 encoder. The temporal model uses a shared frame encoder plus **attention over three frames**, learning changes across time. We evaluate uncertainty via Monte Carlo dropout and calibration metrics.

> Temporal context improves the best final model and gives better-calibrated predictions.

**STATIC MODEL**

t−10 · t−5 · $t_0$

↓ ResNet-18

Final frame only.

**Ignores evolution.**

**TEMPORAL MODEL**

t−10 → t−5 → $t_0$

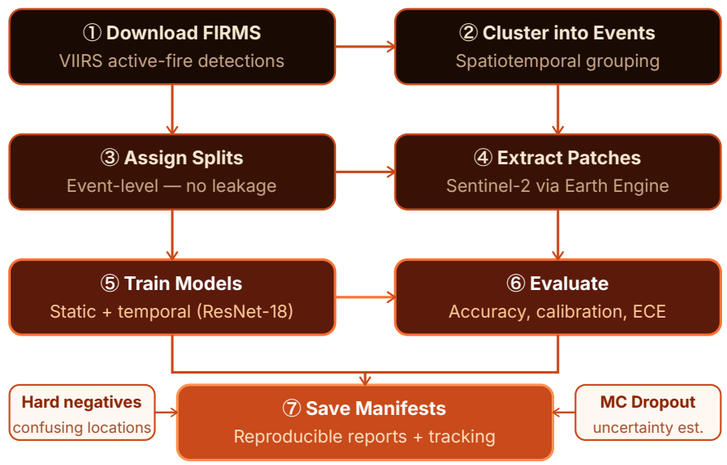
↓ Encoder + Attention

All 3 frames.

**Learns change.**

*Static model uses only $t_0$ frame; temporal model attends over t−10 → t−5 → $t_0$ to learn fire evolution patterns.*

## SYSTEM DESIGN — END-TO-END PIPELINE




Complete end-to-end reproducible pipeline

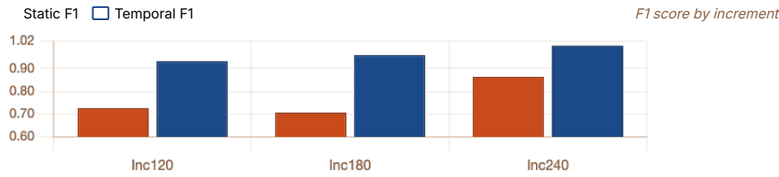
*FIRMS detections → event clustering → split-safe partitioning → Sentinel-2 extraction → dual model training → calibrated evaluation → reproducibility manifests.*

## KEY DESIGN DECISIONS

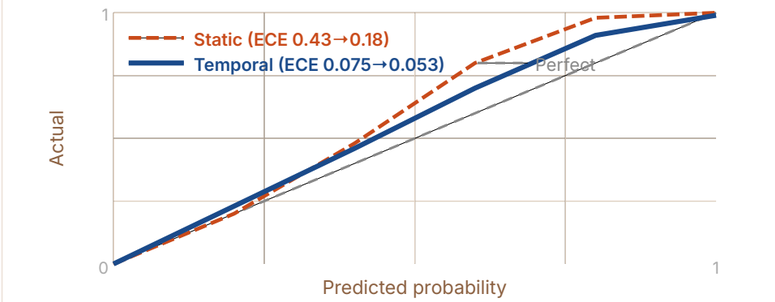
- **Event-level splits** — prevents any leakage between train/val/test
- **Short sequences** (3 frames) — keeps pipeline lightweight and deployable
- **Hard negatives** — confusing nearby locations, reflects real operational noise
- **Reproducibility manifests** — every run is traceable and comparable

## RESULTS — INCREMENTAL COMPARISON

| Model | Split | PR-AUC | F1 | Recall | ECE |
|---|---|---|---|---|---|
| Static | Test | 0.9821 | 0.9456 | 0.9523 | 0.0326 |
| Temporal | Test | 0.9854 | 0.9489 | 0.9732 | 0.0385 |
| Static | OOD | 0.7125 | 0.4103 | 0.5219 | 0.0638 |
| Temporal | OOD | 0.8923 | 0.7106 | 0.8486 | 0.0923 |




## CALIBRATION — RELIABILITY DIAGRAM




*Temporal model tracks the perfect-calibration diagonal far more closely. Lower ECE = predictions are trustworthy, not just accurate.*

**Key takeaway:** Temporal modeling gives the strongest final run and best calibration at Inc240, while static remains competitive at smaller data scales. Temporal does not win every increment — the result is nuanced, not absolute.

## WHAT WE LEARNED

Temporal information matters, but **not uniformly** at every sample size

**Calibration is as important as accuracy** for operational trust

Careful data pipeline design — event-level grouping, leakage prevention, hard negatives — **materially affects results**

## LIMITATIONS & NEXT STEPS

Default real-data extraction does not yet cover every event. More real-data coverage would improve confidence in final claims. Next steps: **broader extraction, larger-scale validation,** and operational threshold tuning.

> The project is a full pipeline, not just a classifier.